

Algorithm Project

Birthday Cake Candles

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Non-Recursive:

Pseudo-code:

Algorithm birthdayCakeCandles(candles, age)

```
maximum <-candles[0]
```

```
count <-1
```

```
For i <- 1 to age - 1 step 1
```

```
do{
```

```
    If candles[i] > maximum
```

```
        maximum <-candles[i]
```

```
        count <-1
```

```
    Else if candles[i] == maximum
```

```
        count <- count + 1
```

```
}
```

```
Return count
```

Analysis & Complexity:

Steps:

1. Read the number of candles.
2. Store candle heights in an array.
3. Compare each candle with the current maximum.
4. Update maximum and count when needed.
5. Print number of tallest candles.

Time Complexity:

$T(n)=O(n)$

Recursive:

Pseudo-code:

```
Algorithm findMaxAndCount(candles, age, index, maximum, count)
```

```
  If index == age  
    Return
```

```
  If candles[index] > maximum  
    maximum <- candles[index]  
    count <- 1
```

```
  Else if candles[index] == maximum  
    count <- count + 1
```

```
  findMaxAndCount(candles, age, index + 1, maximum, count)
```

```
Algorithm birthdayCakeCandles(candles, age)
```

```
maximum <- candles[0]  
count <- 0
```

```
findMaxAndCount(candles, age, 0, maximum, count)
```

```
Return count
```

Analysis & Complexity:

Steps:

1. Read the number of candles.
2. Store candle heights in an array.
3. Use recursion to compare candle heights.
4. Update maximum height and count.
5. Print number of tallest candles.

Time Complexity:

$T(n)=O(n)$